

Add and manage users with Linux commands

I have multiple tasks in this lab:

- Add a new employee
- Change ownership of a file
- Add the new employee to a new group
- Delete the employee from the system

Situation:

In this situation, a new employee with the user name researcher9 joins an organization. You must add it to the system and continue to manage its access during its stay in the organization.

Here are the steps you will follow: First, you will add a new employee to the system and then to the corresponding primary group. Second, you will make this employee own a file related to a particular project. Third, you will add the new employee to a complementary group. Finally, you will erase the employee from the system.

Solution:

- Add a new employee:

```
analyst@68edfb2faa37:~$ sudo useradd researcher9
analyst@68edfb2faa37:~$ sudo usermod -g research_team researcher9
analyst@68edfb2faa37:~$ □
```

I wrote the `useradd` command to add to the system a new user called a 'researcher9'.

Next, I need to add the new user to the `research_team` group.

I use the `usermod` command and the `-g` option to add researcher9 to the `research_team` group and set it to your primary group.

- Change ownership of a file

```
analyst@68edfb2faa37:~$ sudo chown researcher9 /home/researcher2/projects/project_r.txt
analyst@68edfb2faa37:~$
```

Just to let you know, the `sudo` command used because of this is superuser (root user) activities, also the `chown` command to set researcher9 as the owner of file "project_r.txt."

- Add the new employee to a new group

```
analyst@0ef992c2f859:~$ sudo chown researcher9 /home/researcher2/projects/project_r.txt
analyst@0ef992c2f859:~$ sudo usermod -a -G sales_team researcher9
analyst@0ef992c2f859:~$
```

The `usermod` command with the `-a` and `-G` options to add `researcher9` to the `sales_team` group and set it to your secondary group.

Note: The options for Linux commands distinguish uppercase from lowercase; therefore, you must make sure to use a lowercase `-a` and a `-G` in uppercase.

- Delete the employee from the system

```
analyst@0ef992c2f859:~$ sudo userdel researcher9
userdel: group researcher9 not removed because it is not the primary group of user researcher9.
analyst@0ef992c2f859:~$ sudo groupdel researcher9
analyst@0ef992c2f859:~$
```

In this screenshot I used the `userdel` and `groupdel` commands. On the first line, I ran `sudo userdel researcher9` as the main command to delete the user "researcher9". On the second and third lines, the shell returned a warning saying that the group "researcher" was not removed because it wasn't the primary group of that user. On the fourth line, I typed `sudo groupdel researcher9` to delete that leftover group manually, and since there was no output, the command ran successfully.